Functions of Several Variables & Partial Derivatives

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Domain, Range, and Level Curves**

Verify Clairaut's Theorem on equality of mixed partial derivatives for the function $f(x, y) = \cos(2x - 3y) + x^2y^3$ by computing f_{xy} and f_{yx} directly.

1. Compute f_x and f_{xy}: $f_x = \frac{\partial}{\partial x}[\cos(2x - 3y) + x^2y^3] = -2\sin(2x - 3y) + 2xy^3$. $f_{xy} = \frac{\partial}{\partial y}[-2\sin(2x - 3y) + 2xy^3] = -2\cos(2x - 3y)(-3) + 6xy^2 = 6\cos(2x - 3y) + 6xy^2$. 2. Compute f_y and f_{yx}: $f_y = \frac{\partial}{\partial y}[\cos(2x - 3y) + x^2y^3] = 3\sin(2x - 3y) + 3x^2y^2$. $f_{yx} = \frac{\partial}{\partial x}[3\sin(2x - 3y) + 3x^2y^2] = 3\cos(2x - 3y)(2) + 6xy^2 = 6\cos(2x - 3y) + 6xy^2$. Since f is C^2 on $\mathbb{R}^2$.

2 **[Foundational] • Target: First-Order Partial Derivatives**

Let $f(x, y) = x^3e^{2y} - y\ln(x)$ defined on $x > 0$. Compute the first-order partial derivatives $f_x(x, y)$ and $f_y(x, y)$, and evaluate $f_x(1, 0)$ and $f_y(1, 0)$.

1. Partial Derivative with respect to x (treating y as constant): $f_x(x, y) = \frac{\partial}{\partial x}[x^3e^{2y}] - \frac{\partial}{\partial x}[y\ln x] = 3x^2e^{2y} - \frac{y}{x}$. At $(1, 0)$: $f_x(1, 0) = 3(1)^2e^0 - \frac{0}{1} = 3$. 2. Partial Derivative with respect to y (treating x as constant): $f_y(x, y) = \frac{\partial}{\partial y}[x^3e^{2y}] - \frac{\partial}{\partial y}[y\ln x] = 2x^3e^{2y} - \ln(x)$. At $(1, 0)$: $f_y(1, 0) = 2(1)^3e^0 - \ln(1) = 2 - 0 = 2$.

3 **[Foundational] • Target: Higher-Order Partials and Clairaut's Theorem**

A temperature distribution is given by $T(x, y, z) = \frac{120}{1 + x^2 + 2y^2 + 3z^2}$. At the point $P(1, 1, 1)$, find: (a) the direction of steepest temperature increase, (b) the maximum rate of increase, and (c) the equation of the tangent plane to the isothermal level surface passing through P.

1. Compute the gradient $ablaT$: Let $u = 1 + x^2 + 2y^2 + 3z^2$. At $P(1, 1, 1)$, $u = 1 + 1 + 2 + 3 = 7$. $ablaT = -\frac{120}{u^2}\langle 2x, 4y, 6z\rangle$. At $P(1, 1, 1)$: $ablaT(1, 1, 1) = -\frac{120}{49}\langle 2, 4, 6\rangle = -\frac{240}{49}\langle 1, 2, 3\rangle$. (a) Direction of steepest increase: $\mathbf{u}_{\max} = \frac{ablaT}{||ablaT||} = -\frac{\langle 1, 2, 3\rangle}{\sqrt{1 + 4 + 9}} = -\frac{1}{\sqrt{14}}\langle 1, 2, 3\rangle$. (b) Maximum rate of increase.

4 **[Foundational] • Target: Tangent Planes and Linear Approximations**

Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

To evaluate the double integral $\iint_R 2xy\,dA$ where $R = [0, 2] \times [0, 3]$, we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to y first, then x: $\int_0^2 \int_0^3 2xy\,dy\,dx$ First, integrate with respect to y, treating x as a constant: $\int_0^3 2xy\,dy = \left[xy^2\right]_{y=0}^{y=3} = x(3^2) - x(0^2) = 9x$ Now, integrate the result with respect to x.

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5 [Foundational] • Target: Domain, Range, and Level Curves

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve C is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region D enclosed by C, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

The given vector field is $\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle = \langle x^2, y \rangle$. According to Green's Theorem, we need to compute $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$. First, find the partial derivative of Q with respect to x: $\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(y) = 0$. Next, find the partial derivative of P with respect to y: $\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2) = 0$. Therefore.

6 [Foundational] • Target: First-Order Partial Derivatives

Let $z = f(x, y)$ be a differentiable function, with polar coordinates transformation $x = r\cos\theta$ and $y = r\sin\theta$. Express $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, and prove that: $\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = ||ablaz||^2$.

1. Apply the Multivariable Chain Rule: $\frac{\partial z}{\partial r} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} = z_x\cos\theta + z_y\sin\theta$.
$\frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial \theta} = z_x(-r\sin\theta) + z_y(r\cos\theta) = r(-z_x\sin\theta + z_y\cos\theta)$. 2. Divide by r and square both expression.

7 [Foundational] • Target: Higher-Order Partials and Clairaut's Theorem

Use the method of Lagrange multipliers to find the maximum volume of a rectangular box with faces parallel to the coordinate planes inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

Let the corner of the box in the first octant be (x, y, z) with $x, y, z > 0$. The total volume is $V(x, y, z) = (2x)(2y)(2z) = 8xyz$. The constraint is $g(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$. 1. Set up the Lagrange multiplier system $ablaV = \lambda ablag$: $8yz = \lambda\frac{2x}{a^2}, \quad 8xz = \lambda\frac{2y}{b^2}, \quad 8xy = \lambda\frac{2z}{c^2}$. 2. Multiply the three equations by x, y, z respectively.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Tangent Planes and Linear Approximations

Derive the Gaussian integral $I = \int_{-\infty}^{\infty} e^{-x^2/2}dx = \sqrt{2\pi}$ from first principles by evaluating the double integral $I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}dA$ in polar coordinates.

1. Express I^2 as a double integral: By Fubini's theorem on independent variables:
$I^2 = \left(\int_{-\infty}^{\infty} e^{-x^2/2}\,dx\right)\left(\int_{-\infty}^{\infty} e^{-y^2/2}\,dy\right) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)/2}\,dx\,dy$. 2. Convert to polar coordinates: Let $x = r\cos\theta, y = r\sin\theta$, so $x^2 + y^2 = r^2$ and $dA = r\,dr\,d\theta$. The domain $\mathbb{R}^2$ maps to $0 \le r < \infty$ and $0 \le \theta \le 2\pi$.

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9 **[Intermediate]** • **Target: Domain, Range, and Level Curves**

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\,dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta}\int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\cdot r\,dr\,d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2}\int_0^{\frac{6}{\sin\theta+2\cos\theta}} (r\cos\theta + r\sin\theta)\cdot r\,dr\,d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.

The region R is in the first quadrant, bounded by the x-axis ($y = 0$), the y-axis ($x = 0$), and the line $y = -2x + 6$. When $x = 0$, $y = 6$. When $y = 0$, $2x = 6$, so $x = 3$. The vertices of the region are $(0, 0), (3, 0), (0, 6)$. In polar coordinates, $x = r\cos\theta$ and $y = r\sin\theta$. The line $y = -2x + 6$ becomes $r\sin\theta = -2r\cos\theta + 6$. Rearranging for r, we get $r(\sin\theta + 2\cos\theta) = 6$, so $r = \frac{6}{\sin\theta + 2\cos\theta}$. The first quadrant is defined by.

10 **[Intermediate]** • **Target: First-Order Partial Derivatives**

Evaluate the non-elementary integral $\int_0^1\int_y^1 e^{x^2}dx\,dy$ by reversing the order of integration.

1. Sketch the region of integration D: The given bounds are $y \le x \le 1$ for $0 \le y \le 1$. This describes the triangle bounded by the lines $y = 0$, $x = 1$, and $y = x$. 2. Reverse the order to $dy\,dx$: In terms of vertical cross-sections, x ranges from 0 to 1, and for each fixed x, y ranges from 0 up to the line $y = x$: $\int_0^1\int_0^x e^{x^2}dy\,dx$. 3. Evaluate the inner integral: $\int_0^x e^{x^2}dy = e^{x^2}[y]_0^x = xe^{x^2}$. 4. Evaluate the outer integral: Substitute.

11 **[Intermediate]** • **Target: Higher-Order Partials and Clairaut's Theorem**

Use Stokes' Theorem to evaluate $\oint_C \mathbf{F}\cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y^3, x^3, -z^3\rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1$, oriented counterclockwise when viewed from above. Verify surface independence by comparing the flux of $abla \times \mathbf{F}$ through the flat disk $S_1 : x^2 + y^2 \le 1, z = 1$ and the paraboloid cap $S_2 : z = 2 - (x^2 + y^2)$.

1. Compute $abla \times \mathbf{F}$: $abla \times \mathbf{F} = \left|\mathbf{i}, \mathbf{j}, \mathbf{k}; \partial_x, \partial_y, \partial_z; -y^3, x^3, -z^3\right| = \langle 0 - 0,\ 0 - 0,\ 3x^2 - (-3y^2)\rangle = \langle 0,\ 0,\ 3(x^2 + y^2)\rangle$. 2. Flux through flat disk S_1 ($z = 1$, $\mathbf{n} = \mathbf{k}$): On S_1, $d\mathbf{S} = \mathbf{k}\,dA$. $\iint_{S_1}(abla \times \mathbf{F})\cdot d\mathbf{S} = \iint_D 3(x^2 + y^2)\,dA$. In polar coordinates.

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12 **[Intermediate]** • **Target: Tangent Planes and Linear Approximations**

A particle's position in three-dimensional space is given by the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where t represents time. The velocity of the particle is given by $\mathbf{v}(t) = \mathbf{r}'(t)$. Suppose the velocity components are related by the partial differential equations: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. If at time $t = 0$, the particle is at the origin $(0, 0, 0)$ and its velocity is $\mathbf{v}(0) = \langle 5, -2, 0 \rangle$, determine the value of $\frac{\partial^2 v_x}{\partial t^2}$ at $t = 0$.

We are given the partial differential equations relating the velocity components: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. To find $\frac{\partial^2 v_x}{\partial t^2}$, we differentiate the first equation with respect to t: $\frac{\partial^2 v_x}{\partial t^2} = \frac{\partial}{\partial t}(-2x(t) + 3y(t))$. Using the chain rule, this becomes $\frac{\partial^2 v_x}{\partial t^2} = -2\frac{\partial x}{\partial t} + 3\frac{\partial y}{\partial t}$. Sin.

13 **[Intermediate]** • **Target: Domain, Range, and Level Curves**

Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + y\sin z,\ y^3 + z\cos x,\ z^3 + xy \rangle$ across the closed boundary surface of the solid cylinder $E : x^2 + y^2 \le a^2,\ 0 \le z \le h$.

1. Compute the divergence $abla \cdot \mathbf{F}$: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3 + y\sin z) + \frac{\partial}{\partial y}(y^3 + z\cos x) + \frac{\partial}{\partial z}(z^3 + xy) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$. Notice that all cross-coupling transcendental terms vanish identically. 2. Apply the Divergence Theorem: $\iint_{\partial E} \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV$. 3. Convert to cylindrical coordinates: In E, $x^2 + y^2 = r^2$, $0 \le r \le a$.

14 **[Intermediate]** • **Target: First-Order Partial Derivatives**

Use the Green's Theorem line integral formula for area, $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, to derive the area enclosed by the hypocycloid (astroid) $\mathbf{r}(t) = \langle a\cos^3 t,\ a\sin^3 t \rangle$ for $0 \le t \le 2\pi$.

1. Compute the differentials: $dx = -3a\cos^2 t\sin t\,dt, \qquad dy = 3a\sin^2 t\cos t\,dt$. 2. Evaluate $x\,dy - y\,dx$: $x\,dy = (a\cos^3 t)(3a\sin^2 t\cos t\,dt) = 3a^2\cos^4 t\sin^2 t\,dt$. $y\,dx = (a\sin^3 t)(-3a\cos^2 t\sin t\,dt) = -3a^2\sin^4 t\cos^2 t\,dt$. $x\,dy - y\,dx = 3a^2\cos^2 t\sin^2 t(\cos^2 t + \sin^2 t)\,dt = 3a^2\cos^2 t\sin^2 t\,dt$. 3. Use trigonometric identities: Since $\cos t\sin t = \frac{1}{2}\sin(2t)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Higher-Order Partials and Clairaut's Theorem** • **Cross-Lesson: 4.01 Double Integrals**

Let the function $f(x, y)$ be defined as $f(x, y) = 2x^2 - 3y^2$. Calculate the partial derivative of f with respect to y at the point $(2, 3)$.

The function is given by $f(x, y) = 2x^2 - 3y^2$. To find the partial derivative of f with respect to y, denoted as f_y, we treat x as a constant and differentiate with respect to y. $f_y(x, y) = \frac{\partial}{\partial y}(2x^2 - 3y^2)$ Differentiating the term $2x^2$ with respect to y yields 0, as it is a constant with respect to y. Differentiating the term $-3y^2$ with respect to y yields $-3 \cdot 2y = -6y$. Therefore, the partial derivative of f with respect to y is: $f_y(x, y) = -6y$ Now we n.

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16 **[Challenging]** • **Target: Tangent Planes and Linear Approximations** • **Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives**

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let S be the closed surface of the unit cube defined by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through S is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume V enclosed by S. Calculate this flux by evaluating $\iiint_V abla \cdot \mathbf{F}\, dV$.

The vector field is $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. The divergence of $\mathbf{F}$ is $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$. The volume V enclosed by the unit cube is given by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$. The volume of this cube is $1 \times 1 \times 1 = 1$. Therefore, the flux is given by the triple integral: $\iiint_V abla \cdot \mathbf{F}\, dV = \iiint_V 3\, dV$. Since.

17 **[Challenging]** • **Target: Domain, Range, and Level Curves** • **Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers**

Use Green's Theorem to evaluate $\oint_C (y^2 - 7y)\, dx + (2xy + 2x)\, dy$, where C is the positively oriented boundary of the region D enclosed between the parabolas $y = x^2$ and $x = y^2$.

1. Compute the 2D curl: With $P = y^2 - 7y$ and $Q = 2xy + 2x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (2y + 2) - (2y - 7) = 2 - (-7) = 9$. 2. Intersections of boundary curves: $y = x^2$ and $x = y^2 \Rightarrow x = x^4 \Rightarrow x = 0, 1$. Region D is described by $0 \le x \le 1$ and $x^2 \le y \le \sqrt{x}$. 3. Apply Green's Theorem.

18 **[Challenging]** • **Target: First-Order Partial Derivatives** • **Cross-Lesson: 4.01 Double Integrals**

Use Green's Theorem to evaluate the circulation $\oint_C (3y - e^{\sin x})\, dx + (7x + \sqrt{y^4 + 1})\, dy$, where C is the circle $x^2 + y^2 = 9$ oriented counterclockwise.

1. Identify components and compute 2D curl: With $P = 3y - e^{\sin x}$ and $Q = 7x + \sqrt{y^4 + 1}$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{\partial}{\partial x}\left(7x + \sqrt{y^4 + 1}\right) - \frac{\partial}{\partial y}(3y - e^{\sin x}) = 7 - 3 = 4$. The non-elementary transcendental terms vanish completely. 2. Apply Green's Theorem: The region D is the circular disk of radius $R = 3$.

19 **[Challenging]** • **Target: Higher-Order Partials and Clairaut's Theorem** • **Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives**

Consider the function $f(x, y) = 3x^2 - 5y^3$. Find the partial derivative of f with respect to y evaluated at the point $(2, -1)$, i.e., $f_y(2, -1)$.

To find the partial derivative of $f(x, y) = 3x^2 - 5y^3$ with respect to y, we treat x as a constant and differentiate with respect to y. $f_y(x, y) = \frac{\partial}{\partial y}(3x^2 - 5y^3)$ Since $3x^2$ is a constant with respect to y, its derivative is 0. The derivative of $-5y^3$ with respect to y is $-5 \cdot 3y^{3-1} = -15y^2$. Therefore, $f_y(x, y) = -15y^2$. Now, we evaluate $f_y(x, y)$ at the point $(2, -1)$. $f_y(2, -1) = -15(-1)^2$ $f_y(2, -1) = -15(1)$ $f_y(2, -1) = -15$.

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20 [Challenging] • Target: Tangent Planes and Linear Approximations • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle x, y, z \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

Write $z = -x - y + 3$; $d\mathbf{S} = \langle 1, 1, 1 \rangle \, dA$. The flux is $\iint_D \mathbf{F} \cdot \langle 1, 1, 1 \rangle \, dA = \frac{27}{2}$. Final Answer: $\frac{27}{2}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domain, Range, and Level Curves

Find the domain of the function $f(x, y) = \ln(9 - x^2 - y^2)$.

a. **The interior of the circle** $x^2 + y^2 < 9$ ✓

b. The exterior of the circle $x^2 + y^2 > 9$

c. The closed disk $x^2 + y^2 \leq 9$

d. All of $\mathbb{R}^2$

The argument of the natural logarithm must be strictly positive: $9 - x^2 - y^2 > 0 \Leftrightarrow x^2 + y^2 < 9$, which is the open interior of a circle of radius 3 centered at the origin.

22 [Foundational] • Target: Domain, Range, and Level Curves

What geometric curves are the level curves of $z = x^2 + y^2$ for $c > 0$?

a. **Concentric circles centered at** $(0, 0)$ ✓

b. Lines of slope 1

c. Parabolas opening upward

d. Hyperbolas

Setting $z = c \Rightarrow x^2 + y^2 = c$. For $c > 0$, this is a family of concentric circles of radius $\sqrt{c}$.

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23 [Foundational] • Target: Domain, Range, and Level Curves

What is the range of $g(x, y) = \sqrt{16 - x^2 - y^2}$?

a. $[0, 4]$ ✓

b. $[0, 16]$

c. $[-4, 4]$

d. $[0, \infty)$

When $x^2 + y^2 = 0$, $g(0, 0) = \sqrt{16} = 4$. On the boundary $x^2 + y^2 = 16$, $g = 0$. Since square roots are non-negative, the range is $[0, 4]$.

24 [Foundational] • Target: First-Order Partial Derivatives

Find the partial derivative $f_x(x, y)$ for $f(x, y) = 3x^3y^2 - 5x^2 + 4y^3$.

a. $9x^2y^2 - 10x$ ✓

b. $6x^3y + 12y^2$

c. $9x^2y^2 - 10x + 12y^2$

d. $3x^2y^2 - 10x$

Treat y as a constant: $\frac{\partial}{\partial x}[3x^3y^2] = 3(3x^2)y^2 = 9x^2y^2$; $\frac{\partial}{\partial x}[-5x^2] = -10x$; $\frac{\partial}{\partial x}[4y^3] = 0$. Result: $9x^2y^2 - 10x$.

25 [Foundational] • Target: First-Order Partial Derivatives

Find $f_y(x, y)$ for $f(x, y) = e^{2x}\sin(3y)$.

a. $3e^{2x}\cos(3y)$ ✓

b. $2e^{2x}\sin(3y)$

c. $6e^{2x}\cos(3y)$

d. $-3e^{2x}\cos(3y)$

Treat x as constant: $\frac{\partial}{\partial y}[e^{2x}\sin(3y)] = e^{2x} \cdot 3\cos(3y) = 3e^{2x}\cos(3y)$.

26 [Foundational] • Target: First-Order Partial Derivatives

Evaluate $f_x(1, 2)$ for $f(x, y) = x^2y + 3xy^2$.

a. 16 ✓

b. 28

c. 12

d. 8

$f_x(x, y) = 2xy + 3y^2$. At $(1, 2)$: $f_x(1, 2) = 2(1)(2) + 3(2^2) = 4 + 3(4) = 4 + 12 = 16$.

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27 [Foundational] • Target: Higher-Order Partials and Clairaut's Theorem

What does Clairaut's Theorem state about mixed second-order partial derivatives f_{xy} and f_{yx}?

a. If both f_{xy} and f_{yx} are continuous on an open region, then $f_{xy} = f_{yx}$ ✓

b. Mixed partials are always equal to 0

c. The sum $f_{xy} + f_{yx} = 0$

d. Equality holds only for polynomials

Clairaut's Theorem guarantees that mixed partial derivatives are equal ($f_{xy} = f_{yx}$) provided they are continuous on an open disk.

28 [Foundational] • Target: Higher-Order Partials and Clairaut's Theorem

For $f(x, y) = x^3y^4$, verify Clairaut's Theorem by finding f_{xy}.

a. $12x^2y^3$ ✓

b. $3x^2y^4$

c. $4x^3y^3$

d. $6xy^2$

$f_x = 3x^2y^4$. Then $f_{xy} = \frac{\partial}{\partial y}[3x^2y^4] = 3x^2(4y^3) = 12x^2y^3$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Higher-Order Partials and Clairaut's Theorem

Find f_{xx} for $f(x, y) = x^4 - 2x^2y^3 + y^5$.

a. $12x^2 - 4y^3$ ✓

b. $4x^3 - 4xy^3$

c. $12x^2$

d. $-12x^2y^2$

$f_x = 4x^3 - 4xy^3$. Differentiating again with respect to x: $f_{xx} = 12x^2 - 4y^3$.

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30 [Intermediate] • Target: Tangent Planes and Linear Approximations

What is the equation of the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0)?

a. $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$ ✓

b. $z - z_0 = f_x(x - x_0) - f_y(y - y_0)$

c. $z = f_x x_0 + f_y y_0$

d. $(x - x_0) + (y - y_0) + (z - z_0) = 0$

The linear approximation (tangent plane) formula is $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$.

31 [Intermediate] • Target: Tangent Planes and Linear Approximations

Find the equation of the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.

a. $z - 5 = 2(x - 1) + 4(y - 2) \Leftrightarrow 2x + 4y - z = 5$ ✓

b. $2x + 4y + z = 15$

c. $x + 2y - z = 0$

d. $4x + 2y - z = 5$

$f_x = 2x \Rightarrow f_x(1, 2) = 2$. $f_y = 2y \Rightarrow f_y(1, 2) = 4$. Plane: $z - 5 = 2(x - 1) + 4(y - 2) \Rightarrow z - 5 = 2x - 2 + 4y - 8 \Rightarrow 2x + 4y - z = 5$.

32 [Intermediate] • Target: Tangent Planes and Linear Approximations

What is a normal vector to the surface $z = f(x, y)$ at (x_0, y_0)?

a. $\mathbf{n} = [f_x(x_0, y_0), f_y(x_0, y_0), -1]$ ✓

b. $\mathbf{n} = [f_x, f_y, 1]$

c. $\mathbf{n} = [1, 1, 1]$

d. $\mathbf{n} = [-f_y, f_x, 0]$

Rewriting as $F(x, y, z) = f(x, y) - z = 0$, the gradient $\nabla F = [f_x, f_y, -1]$ forms an outward normal vector.

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33 [Intermediate] • Target: Domain, Range, and Level Curves

Consider the function $f(x, y) = 5x^3 - 2y^2 + 7$. Compute the value of the partial derivative $f_y(2, -3)$.

a. 20

b. **-12 ✓**

c. 12

d. -20

To find the partial derivative of $f(x, y)$ with respect to y, we treat x as a constant. Given $f(x, y) = 5x^3 - 2y^2 + 7$. The partial derivative with respect to y is denoted by f_y. $f_y = \frac{\partial}{\partial y}(5x^3 - 2y^2 + 7)$ Applying the power rule and noting that $5x^3$ and 7 are constants with respect to y: $f_y = 0 - 2(2y) + 0$ $f_y = -4y$ Now, we need to evaluate f_y at the point $(2, -3)$: $f_y(2, -3) = -4(-3)$ $f_y(2, -3) = 12$

34 [Intermediate] • Target: First-Order Partial Derivatives

Evaluate the double integral $\iint_R xy\,dA$ where R is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.

a. 1/12

b. **1/8 ✓**

c. 1/6

d. 1/4

The region of integration R is a triangle with vertices (0,0), (1,0), and (1,1). We can set up the double integral using Fubini's Theorem. Since $y = x$ is the upper bound for y and $x = 1$ is the upper bound for x, we can integrate with respect to y first, then x. The limits for y are from 0 to x, and the limits for x are from 0 to 1. $\iint_R xy\,dA = \int_0^1 \int_0^x xy\,dy\,dx$ First, evaluate the inner integral with respect to y.

35 [Intermediate] • Target: Higher-Order Partials and Clairaut's Theorem

Consider the region R in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\,dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\,dr\,d\theta = \int_0^{\pi/4} A\,d\theta$, find the value of A.

a. **4 ✓**

b. $\pi/4$

c. 16

d. 2

The region R is described in the first quadrant, bounded by the x-axis ($\theta = 0$), the line $y = x$ ($\theta = \pi/4$), and the circle $x^2 + y^2 = 16$ ($r = 4$). The integrand is $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$. In polar coordinates, $x^2 + y^2 = r^2$, so $\sqrt{x^2 + y^2} = r$. Thus, $f(x, y) = \frac{1}{r}$. The differential area element in polar coordinates is $dA = r\,dr\,d\theta$. The double integral in polar coordinates is.

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36 [Intermediate] • Target: Tangent Planes and Linear Approximations

For $f(x, y) = 3x^2 + 4y^2$, find $f_x(4, 5)$.

a. **24** ✓ **b.** 30

c. 12 **d.** 40

Applying Partial Derivatives principles, $f_x = 6x$, so $f_x(4, 5) = 6(4) = 24$. Therefore, the verified correct option is 24.

37 [Intermediate] • Target: Domain, Range, and Level Curves

For $f(x, y) = 7x^2 + 2y^2$, find $f_y(3, 6)$.

a. **24** ✓ **b.** 12

c. 42 **d.** -24

Applying Partial Derivatives principles, $f_y = 4y$, so $f_y(3, 6) = 4(6) = 24$. Therefore, the verified correct option is 24.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: First-Order Partial Derivatives • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.

a. -24 **b.** 4

c. **8** ✓ **d.** -42

Applying Partial Derivatives principles, $f_y = 8y$, so $f_y(-3, 1) = 8(1) = 8$. Therefore, the verified correct option is 8.

39 [Challenging] • Target: Higher-Order Partials and Clairaut's Theorem • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 8x^2 + 9y^2$, find $f_x(-4, -3)$.

a. -48 **b.** **-64** ✓

c. -32 **d.** -54

$f_x = 16x$, so $f_x(-4, -3) = 16(-4) = -64$.

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40 [Challenging] • Target: Tangent Planes and Linear Approximations • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = 4x^2 + 7y^2$, find $f_y(-2, -6)$.

a. -28

b. -42

c. **-84** ✓

d. -16

$f_y = 14y$, so $f_y(-2, -6) = 14(-6) = -84$.

41 [Challenging] • Target: Domain, Range, and Level Curves • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 9x^2 + 8y^2$, find $f_y(3, -5)$.

a. **-80** ✓

b. 48

c. -40

d. 54

$f_y = 16y$, so $f_y(3, -5) = 16(-5) = -80$.

42 [Challenging] • Target: First-Order Partial Derivatives • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 8x^2 + 7y^2$, find $f_x(-4, 2)$.

a. 32

b. -32

c. 28

d. **-64** ✓

$f_x = 16x$, so $f_x(-4, 2) = 16(-4) = -64$.

43 [Challenging] • Target: Higher-Order Partials and Clairaut's Theorem • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = 5x^2 + 6y^2$, find $f_x(-5, 1)$.

a. **-50** ✓

b. 10

c. -25

d. 12

$f_x = 10x$, so $f_x(-5, 1) = 10(-5) = -50$.

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44 [Challenging] • Target: Tangent Planes and Linear Approximations • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 + 7y^2$, find $f_x(-5, -2)$.

a. -16

b. -20

c. -28

d. **-40 ✓**

$f_x = 8x$, so $f_x(-5, -2) = 8(-5) = -40$.

45 [Challenging] • Target: Domain, Range, and Level Curves • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 4x^2 + 3y^2$, find $f_y(-4, -3)$.

a. **-18 ✓**

b. -24

c. -9

d. -32

$f_y = 6y$, so $f_y(-4, -3) = 6(-3) = -18$.

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